RESTful API Assignment Submission Report

Assignment Name:	Build a RESTful API using Node.js and Express (100 Marks)
Project Folder:	express-rest-assignment
Tech Stack:	Node.js, Express.js, Thunder Client / Postman
Data Source:	In-Memory Array (as per assignment requirements)

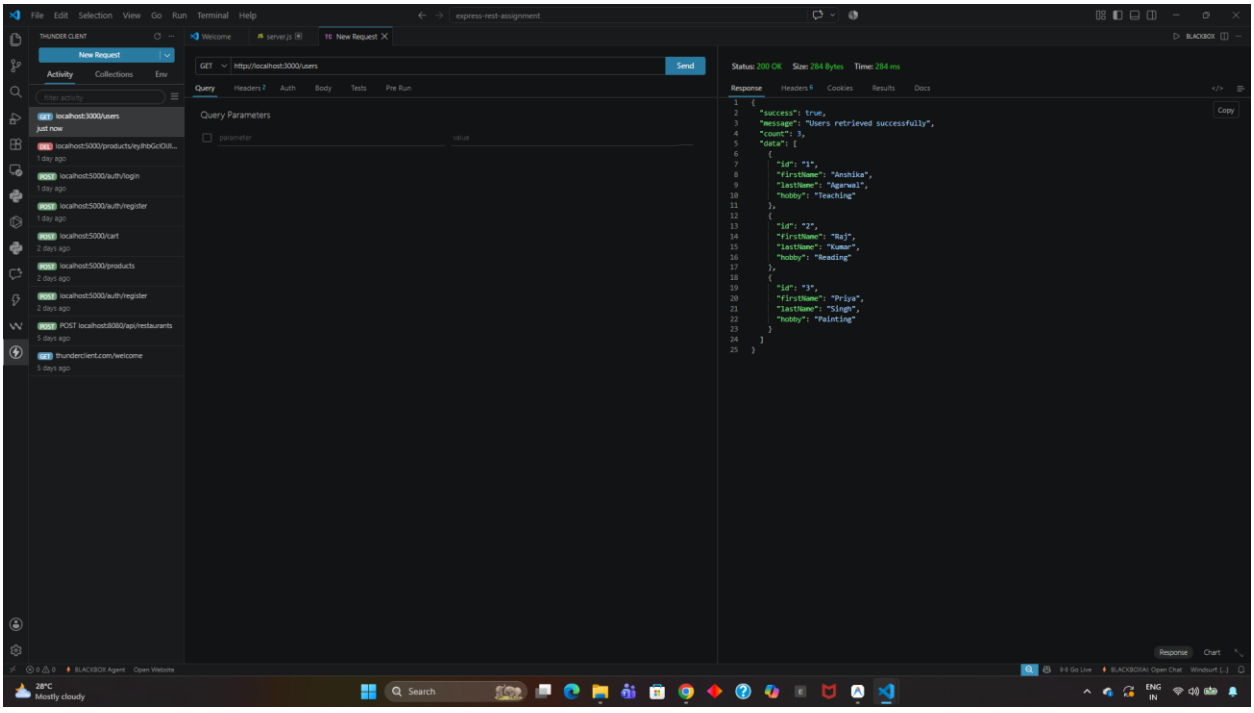
1. Overview

This report documents the implementation and testing of a RESTful API built using Node.js and Express. The API provides full CRUD operations for managing user records in an in-memory array. It incorporates logging middleware, request validation middleware, error handling, and standard HTTP status codes.

2. API Testing Screenshots & Verification

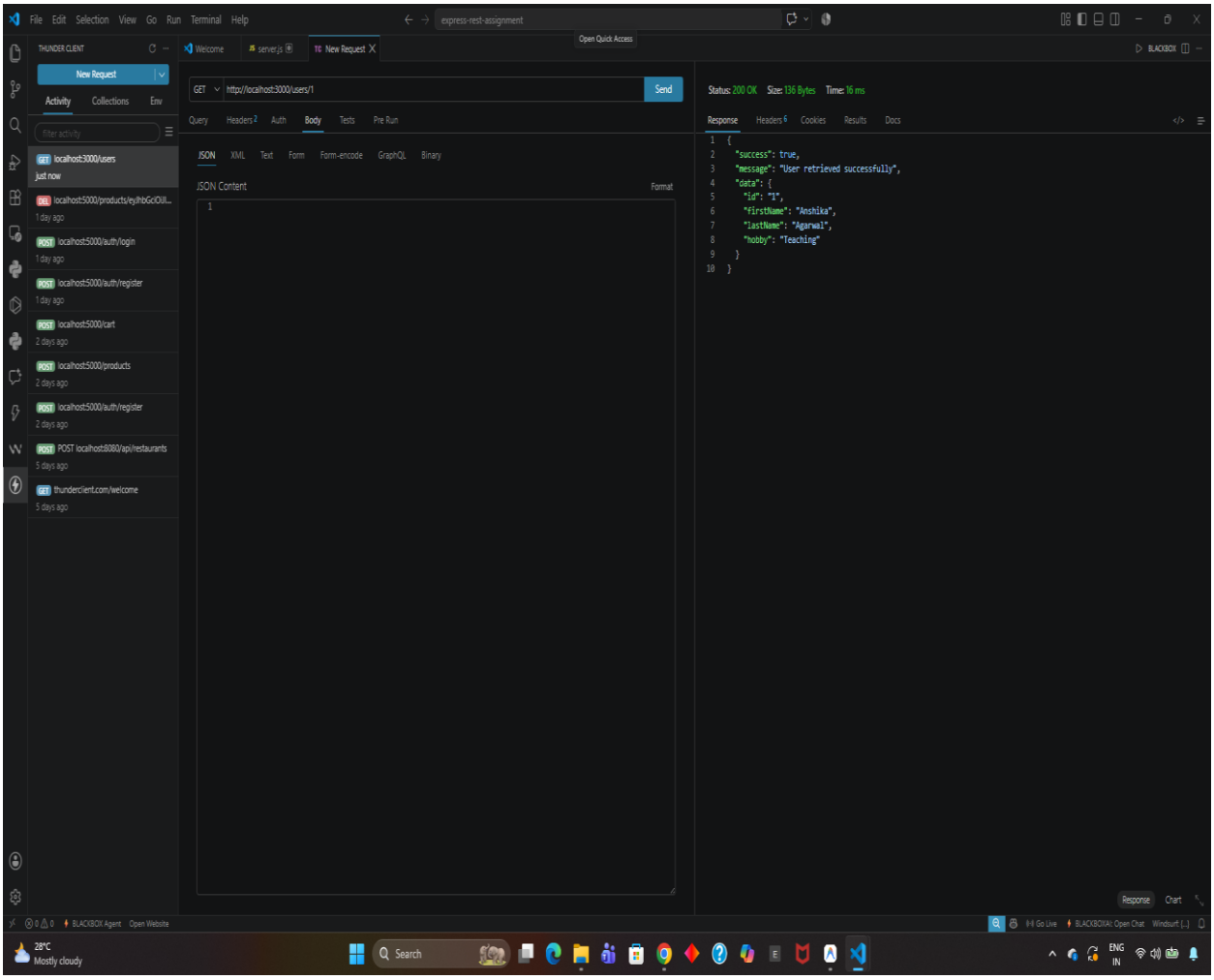
Test 1: GET /users (Fetch All Users)

URL: http://localhost:3000/users
Method: GET | **Status Code:** 200 OK



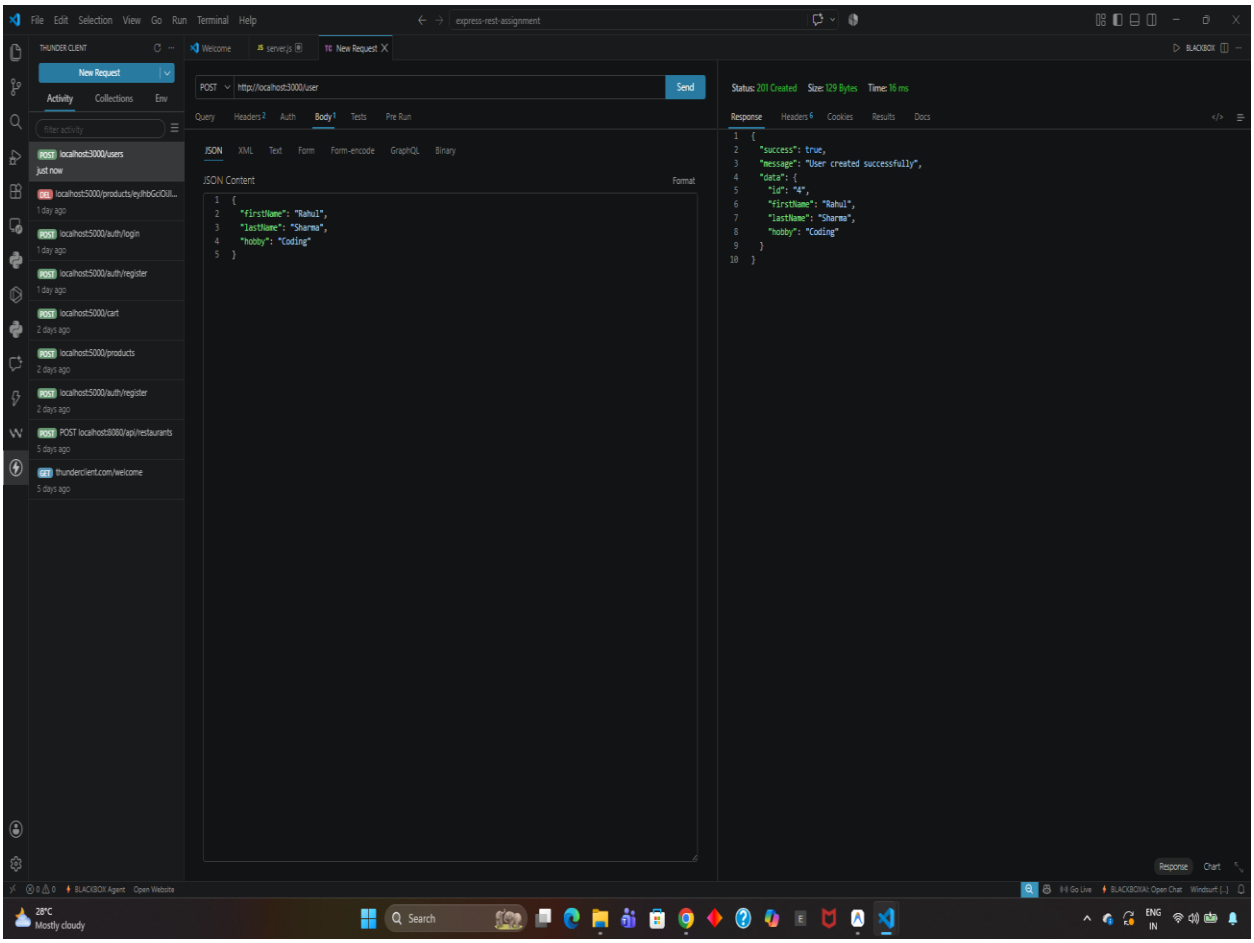
Test 2: GET /users/:id (Fetch User by ID)

URL: http://localhost:3000/users/1
Method: GET | **Status Code:** 200 OK



Test 3: POST /user (Add New User - Success)

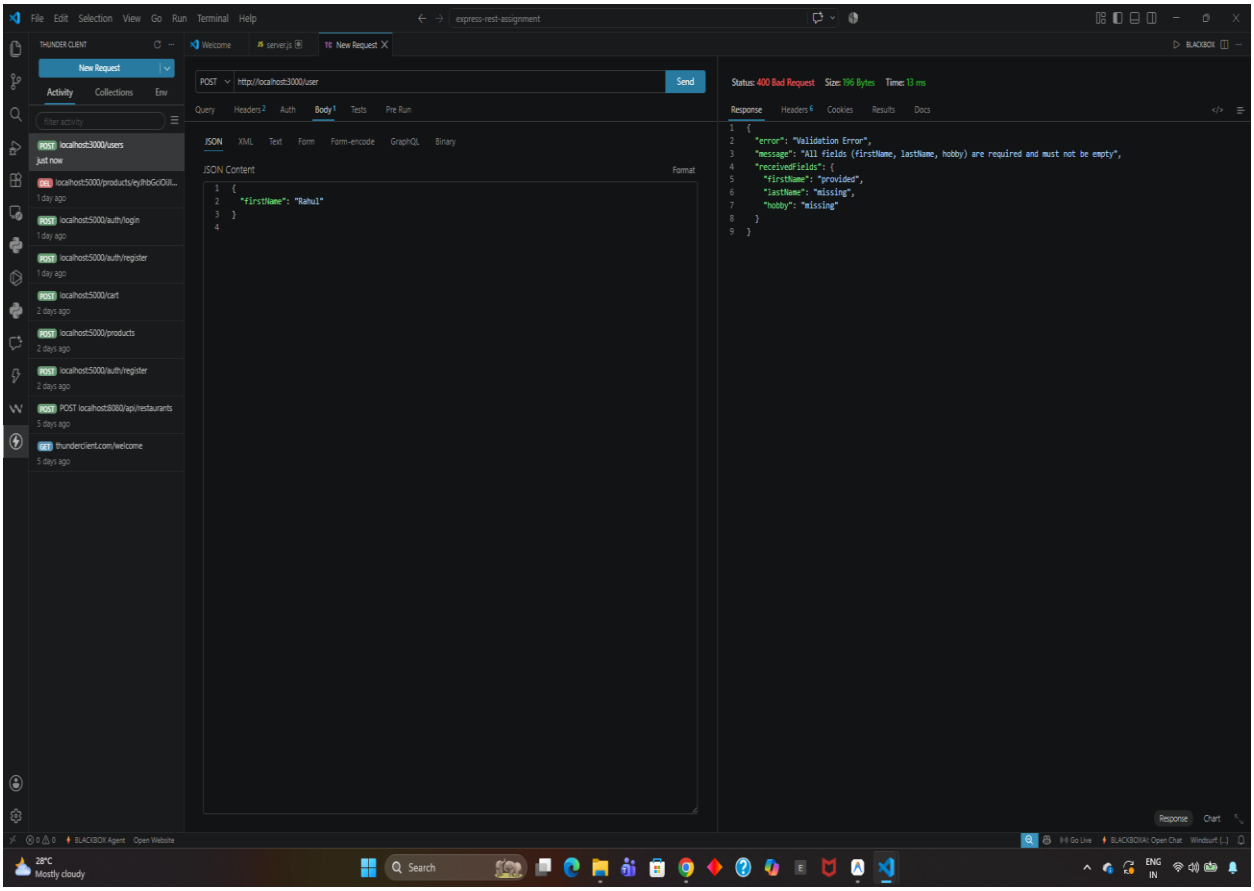
URL: http://localhost:3000/user
Method: POST | **Status Code:** 201 Created


File Edit Selection View Go Run Terminal Help
express-rest-assignment
THUNDER CLIENT
New Request
Activity
Collections
Env
Filter activity
POST localhost:3000/users
just now
DEL localhost:5000/products/eyJhbGciOiJI...
1 day ago
POST localhost:5000/auth/login
1 day ago
POST localhost:5000/auth/register
1 day ago
POST localhost:5000/cart
2 days ago
POST localhost:5000/products
2 days ago
POST localhost:5000/auth/register
2 days ago
POST POST localhost:8080/api/restaurants
5 days ago
GET thunderclient.com/welcome
5 days ago
Welcome
server.js
New Request
BLACKBOX
POST
http://localhost:3000/user
Send
Query
Headers 2
Auth
Body 1
Tests
Pre Run
JSON
XML
Text
Form
Form-encode
GraphQL
Binary
JSON Content
Format
{
"firstName": "Rahul",
"lastName": "Sharma",
"hobby": "Coding"
}
Status: 201 Created
Size: 129 Bytes
Time: 16 ms
Response
Headers 6
Cookies
Results
Docs
{
"success": true,
"message": "User created successfully",
"data": {
"id": "4",
"firstName": "Rahul",
"lastName": "Sharma",
"hobby": "Coding"
}
}
Response
Chart
BLACKBOX Agent
Open Website
Go Live
BLACKBOX: Open Chat
Windsurf (...)
28°C
Mostly cloudy
Search
ENG
IN

Test 4: POST /user (Validation Error Case)

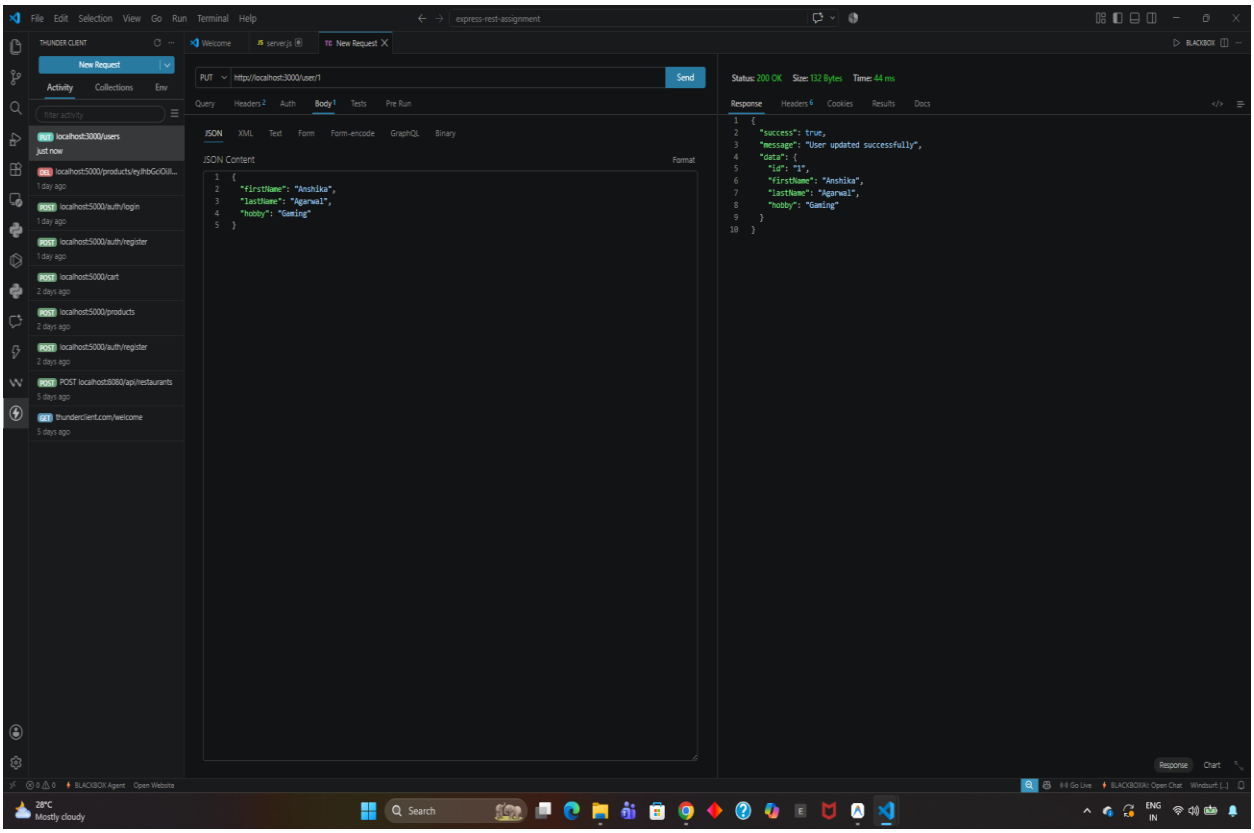
URL: http://localhost:3000/user

Method: POST | **Status Code:** 400 Bad Request



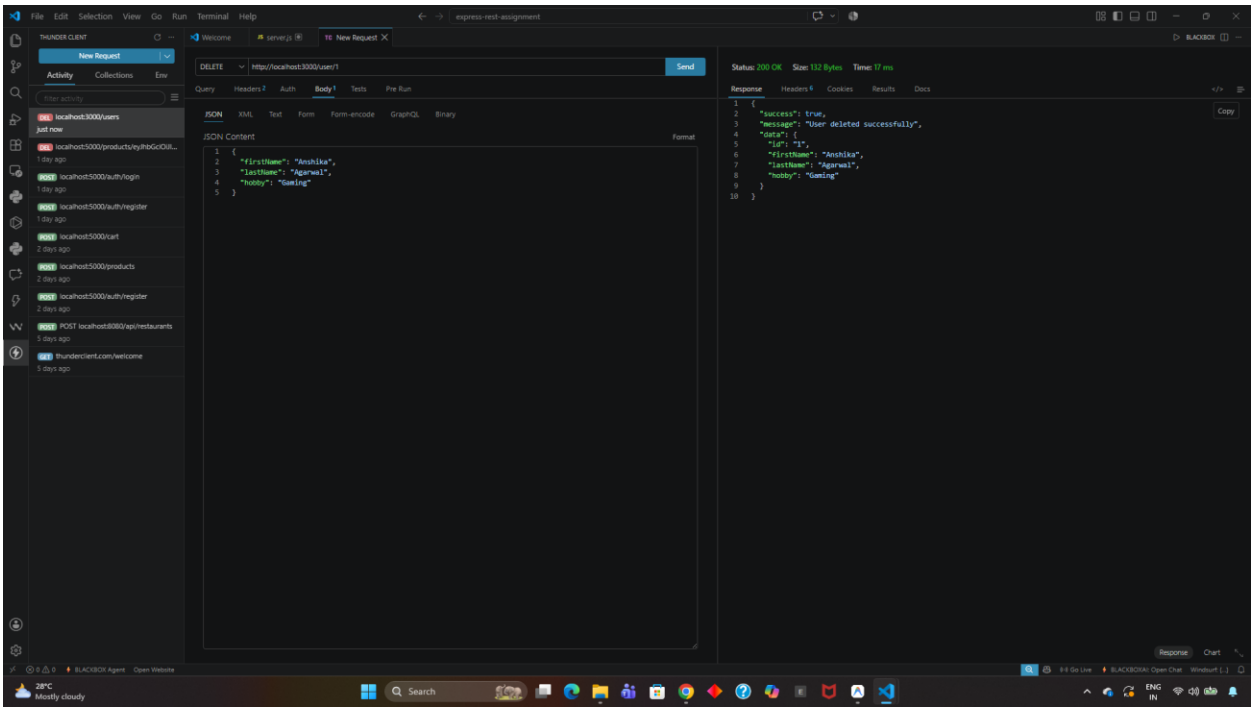
Test 5: PUT /user/:id (Update User Details)

URL: http://localhost:3000/user/1
Method: PUT | **Status Code:** 200 OK



Test 6: DELETE /user/:id (Delete User)

URL: http://localhost:3000/user/1
Method: DELETE | **Status Code:** 200 OK



3. Conclusion

All API endpoints have been tested thoroughly using Thunder Client. The API handles CRUD operations properly, enforces data validation, logs requests, and responds with appropriate HTTP status codes.